

Manifestations of Sick Cell Disorder at Abdominal and Pelvic Imaging

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Abbreviations: ACS = acute chest syndrome, EMH = extramedullary hematopoiesis, ESRD = end-stage renal disease, Hb = hemoglobin, HbS = sickle Hb, HbSS = homozygous sickle cell hemoglobin, HCC = hepatocellular carcinoma, RBC = red blood cell, RMC = renal medullary carcinoma, SCD = sickle cell disorder, VOC = vascular occlusion

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Sickle cell disorder (SCD) refers to a spectrum of hematologic disorders that cause a characteristic clinical syndrome affecting the entire body. It is the most prevalent monogenetic hemoglobinopathy worldwide, with a wide range of focal and systemic expressions. Hemoglobin gene mutation leads to the formation of abnormal sickle-shaped red blood cells, which cause vascular occlusion and result in tissue and organ ischemia and infarction. Recurrent episodes of acute illness lead to progressive multisystem organ damage and dysfunction. Vaso-occlusion, hemolysis, and infection as a result of functional asplenia are at the core of the disease manifestations. Imaging plays an essential role in the diagnosis and management of SCD-related complications in the abdomen and pelvis. A thorough understanding of the key imaging findings of SCD complications involving hepatobiliary, gastrointestinal, genitourinary, and musculoskeletal systems is crucial to timely recognition and accurate diagnosis. The authors aim to familiarize the radiologist with the SCD spectrum, focusing on the detection and evaluation of manifestations that may appear at imaging of the abdomen and pelvis. The topics the authors address include (a) the pathophysiology of the disease, (b) the placement of SCD among hemoglobinopathies, (c) the clinical presentation of SCD, (d) the role of imaging in the evaluation and diagnosis of patients with SCD who present with abdominal and pelvic manifestations in addition to extraperitoneal manifestations detectable at abdominal or pelvic imaging, (e) imaging features associated with common and uncommon sequelae of SCD in abdominal and pelvic imaging studies, and (f) a brief overview of management and treatment of patients with SCD.

Online supplemental material is available for this article.

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the pathophysiology of hemoglobinopathies and RBC injury in SCD.
- Define the imaging features of the common manifestations of SCD in the abdomen and pelvis.
- Review the role of imaging in the diagnosis and evaluation of suspected complications of SCD in the abdomen and pelvis.

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Introduction

Sickle cell disorder (SCD) refers to a genetic disease spectrum characterized by vascular occlusion (VOC) by abnormally shaped (sickled) red blood cells (RBCs) with recurrent episodes of tissue and organ ischemia and infarction leading to progressive multisystem organ damage (1,2). Hemoglobinopathies (inherited disorders

TEACHING POINTS

- Microvascular obstruction by abnormally shaped erythrocytes results in painful tissue ischemia, with subsequent restoration of blood flow causing reperfusion injury. This recurring series of events, leading to progressive ischemic organ damage and chronic complications, characterizes SCD.
- In SCD, hemoglobin production increases in the setting of chronic hemolytic anemia, occurring both within (intramedullary) and outside (extramedullary) the bone marrow. In the abdomen and pelvis, imaging findings of extramedullary hematopoiesis (EMH) include a focal soft-tissue mass(es) within solid organs (the liver and spleen are most commonly affected); organ enlargement; or a soft-tissue mass(es) in the perirenal space, omentum, small bowel mesentery, peritoneum, paravertebral region, thorax, or even skin.
- In SCD, increased polymerization of HbS in the renal medulla leads to VOC of the vasa recta and subsequent ischemia, infarction, papillary necrosis, and scarring, any of which can progress to acute failure.
- Cholestasis and cholangiopathy occur when recurrent endarterial and hepatic sinusoidal obstruction results in ischemic damage to the biliary tree and hepatotoxicity from disruption of biliary flow.
- Infarction most commonly occurs in medullary cavities and epiphyses of the bones, often demonstrates serpiginous regions of sclerosis, and can be associated with subperiosteal hemorrhage. Central growth plate infarction in the spine can result in endplate depressions of the vertebral bodies, resulting in H-shaped vertebrae and adjacent "tower vertebrae" with visible growth deformities.

of hemoglobin [Hb] synthesis), which include SCD, are the most prevalent monogenetic disorders worldwide (3,4). In SCD, Hb synthesis is disrupted by a gene mutation, resulting in the formation of sickle Hb (HbS) (4) (Fig 1). The SCD spectrum is the most prevalent hemoglobinopathy, but α - and β -thalassemia are also common and may co-occur (Fig 2) (1–5).

Globally, approximately 300 000 infants with SCD are born each year, and the number of patients with SCD is expected to increase owing to declines in infant mortality and improvements in life expectancy (3,5). Ninety percent of patients with SCD live in low- and middle-income countries, with the highest prevalence in sub-Saharan Africa, the Mediterranean basin, the Middle East, and the Indian subcontinent (3–6).

Pathophysiology and Clinical Presentation

An HbS mutation affects molecular stability and RBC function (1,3,6). Deoxygenated HbS polymerizes, resulting in Hb aggregation and clumping within the cell. This forms a rigid elongated sickle-shaped RBC, further reducing HbS's affinity for oxygen (2,6). While polymerization can be a reversible process with resolution in the oxygenated state, it can also lead to RBC dehy-

dration by activation of several cell membrane transport systems resulting in "dense" RBCs (an irreversible process) (6).

The mechanisms behind SCD manifestations are complex and multifactorial (Fig 3) (7). Ultimately, microvascular obstruction by abnormally shaped erythrocytes results in painful tissue ischemia, with subsequent restoration of blood flow causing reperfusion injury (1,2). This recurring series of events, leading to progressive ischemic organ damage and chronic complications, characterizes SCD (3).

By age 1 year, children may develop anemia, pain crises, stroke, and widespread organ damage, with the highest mortality rates occurring between 6 months and 3 years of age (5,8). The earliest signs and symptoms of disease frequently include jaundice (icterus), fatigue, fussiness, and dactylitis (painful swelling in the hands and feet) (9). Stunted growth and delayed puberty are common (9). Splenic dysfunction increases the risk of infection, with pneumococcal sepsis being the leading cause of death among children (5,8).

In this article, we focus specifically on the abdominal and pelvic manifestations of SCD as seen at various imaging modalities (Fig 4).

Vascular Manifestations

Vascular manifestations of SCD include vasculopathy, VOC, and acute pain crises (2). Vasculopathies are attributable to increased viscosity and decreased deformability of abnormal RBCs (Fig 3). Additionally, increased affinity of RBCs to vascular endothelium may worsen VOC. Risk factors include hypoxia, dehydration, infections, fever, high altitude, prolonged immobility, extreme temperatures, physical or emotional stress, trauma, steroid therapy, and anesthetics (2). While small-vessel occlusion is typically below the resolution of cross-sectional imaging, evidence of infarction may be seen (7). Large-vessel vasculitis can cause vessel narrowing with or without inflammatory stranding (7). Treatment includes hydration, pain control, proton pump inhibitors, and, in the absence of neurologic symptoms, anticoagulation therapy (2).

Multiorgan Manifestations

Iron Deposition

In chronic blood transfusion therapy, iron accumulation exceeds iron-binding capacity, leading to overload in reticuloendothelial cells (hemosiderosis) including spleen, bone marrow, and liver cells (2,7). In contrast to many inherited anemias, such as β -thalassemia major, iron overload in SCD does not occur without blood transfusion; thus, a different pattern of iron deposition is

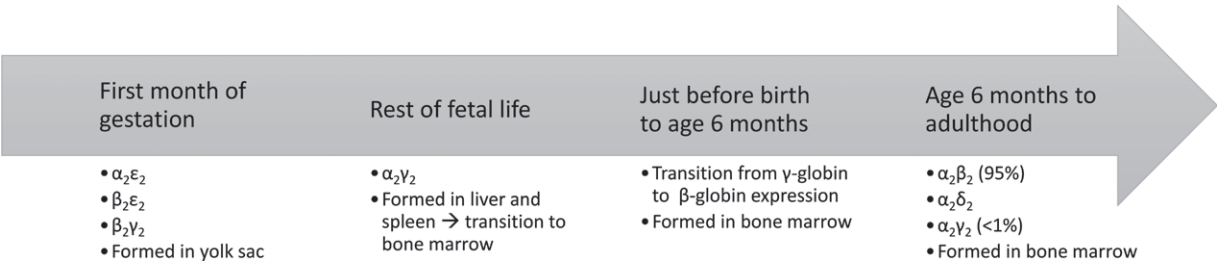


Figure 1. Normal course of Hb synthesis (4). The normal adult human Hb molecule (HbA, not mutated) consists of four globin chains: two α chains and two β chains. When α and β chains are normal, the Hb molecule is abbreviated as HbAA. $\alpha_2, \epsilon_2, \beta_2, \gamma_2,$ and δ_2 are various globin chains. HbF is the fetal hemoglobin molecule.

Sickle cell anemia	Hemoglobin SC disease	HbS/ β -thalassemia	Sickle cell trait
• HbS homozygous inheritance → HbSS • Most common type in those with African origins	• HbS and HbC co-inheritance • Lysine substituted for glutamic acid → HbC • Milder disease	• β-thalassemia co-inheritance • HbS/β^0-thalassemia → severe disease • HbS/β^+-thalassemia less affected	• HbS and HbA co-inheritance • RBC sickling does not tend to occur

Figure 2. Characterization of the most common genotypes of SCD (1,2,5,6). When the β -globin gene is mutated on both chains, a homozygous sickle cell Hb molecule (HbSS) is produced, resulting in SCD. When the sickle cell hemoglobin gene is present only on one β chain (HbS) and the second β -globin gene is normal (HbA), a benign condition termed *sickle cell trait* results (HbSA). In contrast to HbSS, HbSA is stable and is usually not associated with vaso-occlusive disease. At least 10 additional genotypes causing SCD have been described (3). Except for HbS/ β^0 -thalassemia (in which HbA is not produced), compound heterozygosity is usually less severe (3).

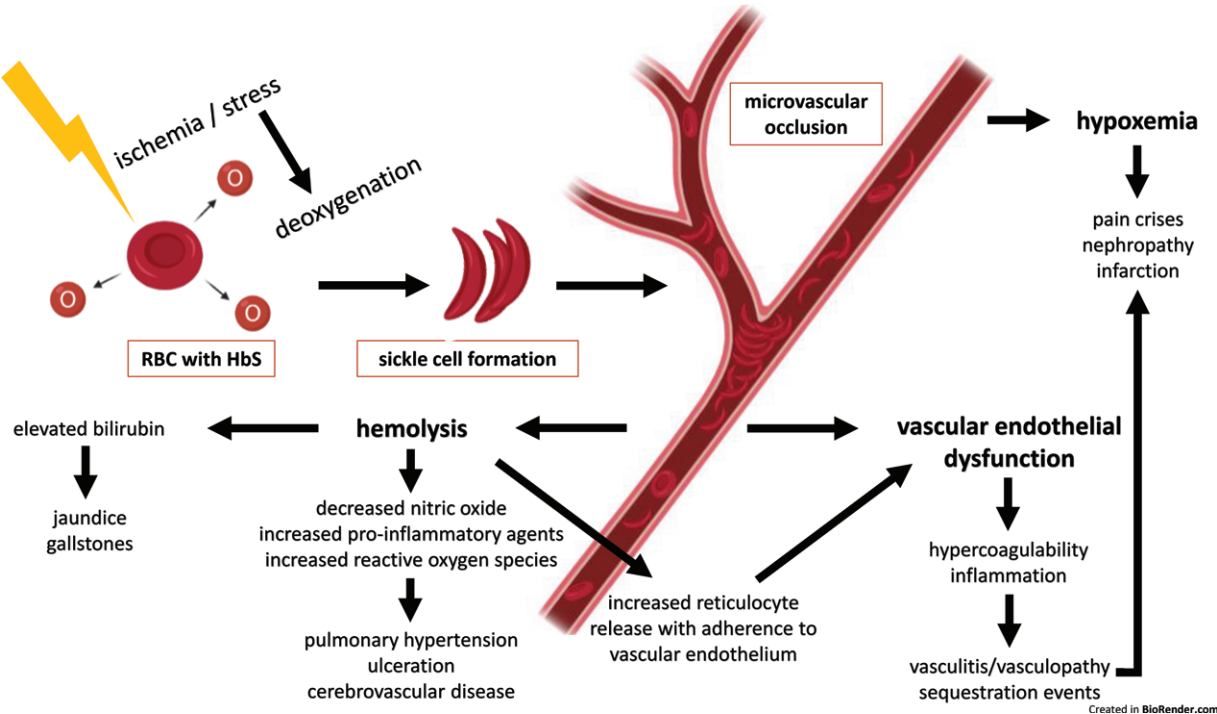


Figure 3. Diagram shows the pathophysiology of the manifestations of SCD (2,6,7,11).

seen. Because transfusion therapy for SCD tends to commence later in life, glandular iron deposition is less common (Table 1) (2,6,10).
US images demonstrate increased echogenicity of solid organs, particularly the liver, spleen, and

kidneys, while nonenhanced CT images demonstrate increased organ size and attenuation (up to 75–135 HU) (Fig E1) (7,11). Signs of liver cirrhosis and portal hypertension can be seen at later stages of iron deposition (Fig E2).

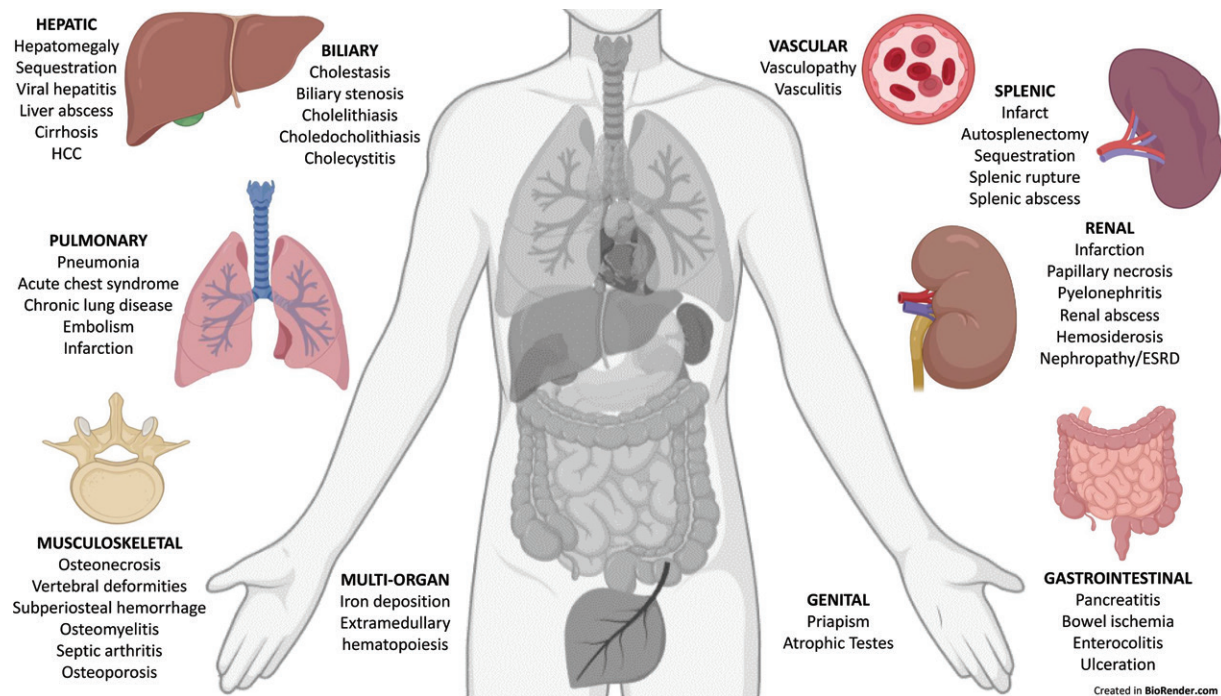


Figure 4. Overview of abdominal and pelvic manifestations of SCD. ESRD = end-stage renal disease, HCC = hepatocellular carcinoma.

MR images show diffuse low signal intensity of the affected organs and bone marrow on T1- and T2-weighted images and signal loss at in-phase imaging compared with opposed-phase imaging owing to the magnetic susceptibility of hemosiderin (Fig 5) (7,12,13). Hemolysis results in deposition of hemosiderin in the renal proximal convoluted tubules, which appears as low signal intensity of the renal cortex at T1-weighted imaging relative to the medulla and decreased signal intensity at T2-weighted imaging.

Because high concentrations of iron can be toxic to the organs, especially the liver, detection and grading of iron overload is critical. Iron deposition causes local field inhomogeneities resulting in T2* shortening at heavily T2-weighted imaging. The degree of T2* shortening closely correlates with the degree of iron deposition. Thus, T2* values less than 10 msec indicate severe iron deposition, 10–20 msec indicate mild-to-moderate deposition, and more than 20 msec denote a normal range. Noninvasive quantification of iron with MRI (heavily T2-weighted fast spin-echo or T2-weighted gradient-echo [GRE] or multiecho sequences) is now considered the standard of care in diagnosis and monitoring of iron overload diseases (14).

Extramedullary Hematopoiesis

In SCD, hemoglobin production increases in the setting of chronic hemolytic anemia, occurring both within (intramedullary) and outside (extramedullary) the bone marrow (15,16). In the abdomen

Table 1: Characteristics of Transfusion-related Iron Deposition in SCD and in Diseases Characterized by Iron Deposition and Transfusion-dependent Thalassemia*

Transfusion-related iron deposition in SCD

- Later onset, lower frequency of transfusion therapy
- Mainly solid organs affected (fewer effects on growth and sexual development)
- Renal cortical iron deposition only seen when SCD hemolytic crisis causes intravascular hemolysis
- Pancreas is usually spared
- Maintenance of effective erythropoiesis

Iron-deposition diseases and transfusion-related thalassemia

- Earlier onset, higher frequency of transfusion therapy
- More diffuse deposition
- Solid organs and endocrine glands (affects growth and sexual development)
- Commonly associated with ineffective erythropoiesis
- Crucial role in iron storage and deposition

* In both cases, iron overload is mitigated by iron chelation (2,6,11).

and pelvis, imaging findings of extramedullary hematopoiesis (EMH) include a focal soft-tissue mass(es) within solid organs (the liver and spleen are most commonly affected); organ enlargement; or a soft-tissue mass(es) in the perirenal space,

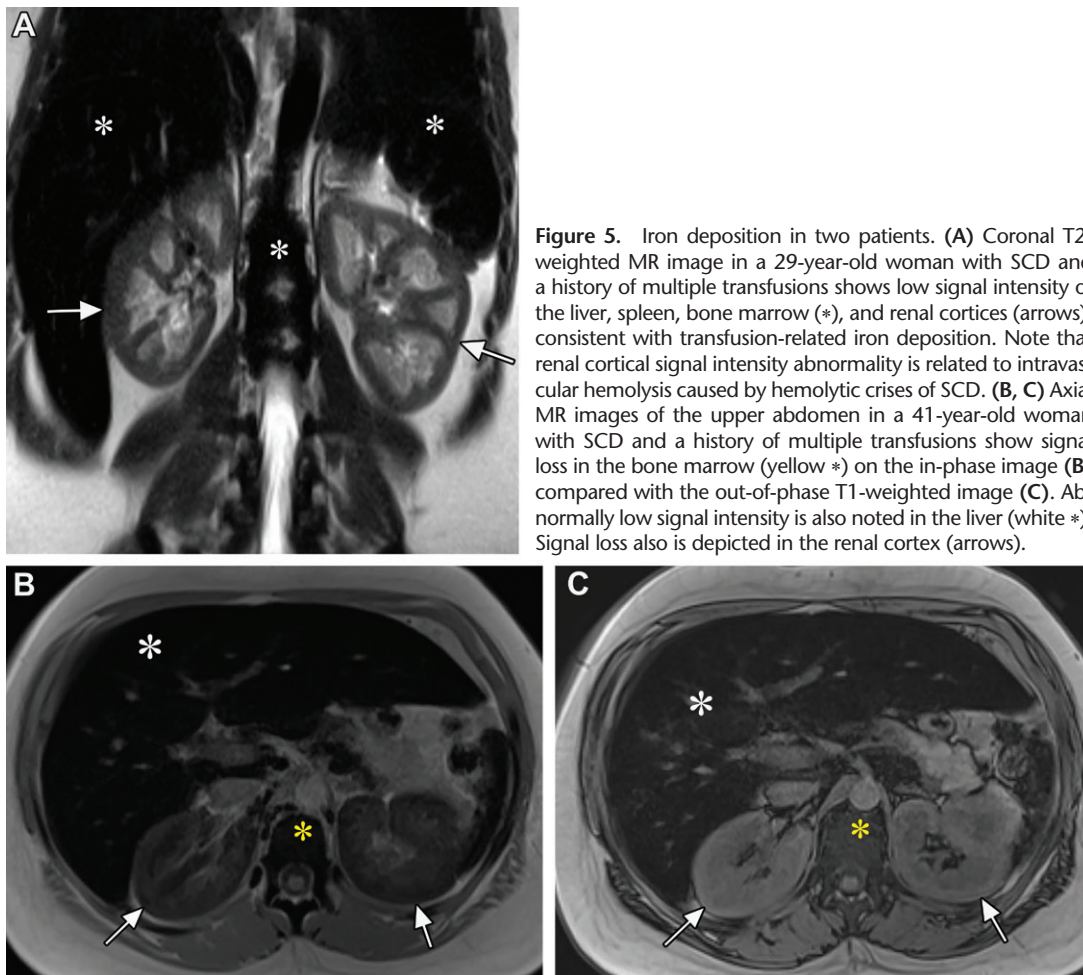


Figure 5. Iron deposition in two patients. (A) Coronal T2-weighted MR image in a 29-year-old woman with SCD and a history of multiple transfusions shows low signal intensity of the liver, spleen, bone marrow (*), and renal cortices (arrows), consistent with transfusion-related iron deposition. Note that renal cortical signal intensity abnormality is related to intravascular hemolysis caused by hemolytic crises of SCD. (B, C) Axial MR images of the upper abdomen in a 41-year-old woman with SCD and a history of multiple transfusions show signal loss in the bone marrow (yellow *) on the in-phase image (B) compared with the out-of-phase T1-weighted image (C). Abnormally low signal intensity is also noted in the liver (white *). Signal loss also is depicted in the renal cortex (arrows).

omentum, small bowel mesentery, peritoneum, paravertebral region, thorax, or even skin (Figs 6, E3) (7,16,17). EMH masses may be solitary, multiple, or separate or contiguous and are slow-growing with well-defined borders (13). At MRI, active hematopoietic tissue has intermediate signal intensity on T1-weighted images and isointensity or high signal intensity on T2-weighted images. Hyperintense signal on T1-weighted and T2-weighted images could be due to fat replacement. At postcontrast T1-weighted imaging, EMH usually demonstrates mild enhancement. Nonfunctional hematopoietic tissue usually shows no postcontrast enhancement and low signal intensity on both T1-weighted and T2-weighted images (7,13). The presence of iron in EMH masses usually results in reduced signal intensity on in-phase T1-weighted GRE images compared with opposed-phase images (13).

Splenic Manifestations

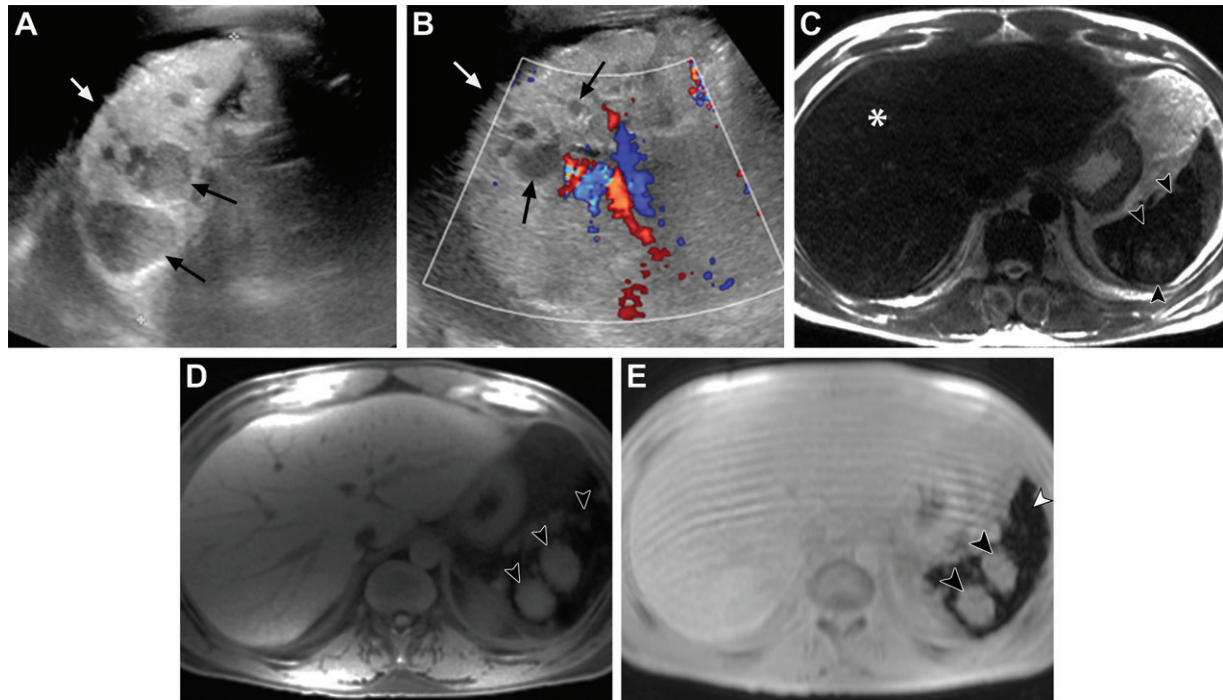
Splenomegaly and Splenic Sequestration

Splenomegaly occurs in approximately 10% of patients older than 10 years and results from vascular congestion (including sequestration),

infarction, or abscess (7,11). At CT, splenomegaly is characterized by a craniocaudal diameter greater than about 12 cm (Fig 7) (7).

Sequestration is characterized by rapid pooling of blood in a solid organ, most commonly the spleen, followed by the liver. In acute sequestration syndrome, splenic trapping of sickled RBCs leads to acute vascular congestion and anemia (2,7,11). Occurring mainly in children with homozygous sickle cell Hb (HbSS) genotype, sequestration may be triggered by pain crisis, infection, or acute chest syndrome (ACS) and complicated by cardiovascular collapse, splenic rupture and/or hemorrhage, and potentially death (2,11,18). Sequestration occurs in 10%–30% of children ages 6 months to 3 years and in approximately 13% of patients overall (although rare in patients older than 8 years). Approximately 50% of children experience recurrent episodes (7,11). The clinical presentation includes sudden weakness or lethargy, tachycardia, abdominal pain, and pallor (7,11). Images show acute splenic enlargement or larger-than-expected size for an autoinfarcted spleen with or without intraparenchymal or

Figure 6. Extramedullary hematopoiesis in a 20-year-old man with SCD who presented with fever and concern for abscess. (A, B) Gray-scale (A) and color Doppler (B) US images show a small echogenic spleen with multiple hypoechoic splenic masses (black arrows) demonstrating perivascular flow. Note the small left pleural effusion (white arrow). (C, D) Axial T2-weighted (C) and gadolinium-enhanced T1-weighted (D) correlative MR images show iron deposition of the liver (* in C) and spleen, with multiple T2 slightly hyperintense and gadolinium-enhancing masses (arrowheads). (E) Axial T1-weighted in-phase image shows signal loss (white arrowhead) in the spleen consistent with iron deposition. No significant signal loss is seen within the masses (black arrowheads). Note that the liver and pancreas are not hypointense like the spleen.



perisplenic hemorrhage (Fig 8) (7,11). On US images, the enlarged spleen may appear homogeneously hypoechoic or heterogeneous, with hypoechoic areas at the periphery signifying infarcts (7).

Peripheral splenic infarcts are more conspicuous at other types of cross-sectional imaging and are characterized by areas of hypoenhancement (7,11). On T1-weighted images, areas of high signal intensity represent intraparenchymal hemorrhage (19). The presence of complex perisplenic and/or peritoneal fluid should raise concern for splenic rupture. Treatment includes urgent simple transfusion (blood is administered but none is removed), with antibiotic therapy in the setting of fever (2,18). Splenectomy is reserved for patients with recurrent sequestration crises, hypersplenism, massive infarction, or splenic abscess (2).

Splenic Infarction and Autosplenectomy

In SCD, recurrent VOC and subsequent splenic infarcts lead to functional asplenia and autosplenectomy, usually by age 5 years and often within the first 18–36 months of life (2,6,7,11). Repeated infarcts lead to splenic shrinking, with replacement of healthy tissue by fibrosis, calcium, and hemosiderin deposition (Fig 9) (11). At contrast-enhanced CT, splenic infarcts appear as

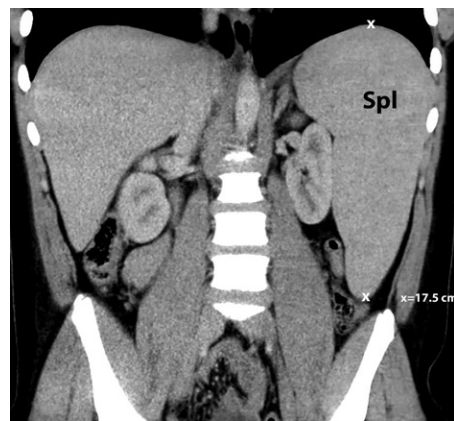


Figure 7. Splenomegaly in a 14-year-old boy with HbSC disease. Coronal intravenous contrast-enhanced CT image shows marked splenomegaly, with the spleen (Spl) measuring 17.5 cm in the craniocaudal dimension.

wedge-shaped peripheral areas of hypoenhancement and may be associated with splenomegaly in the acute phase (Fig 10) (11). On color Doppler US images, infarcted tissue is hypoechoic with no flow within. On MR images, infarcts appear with decreased signal intensity on T1- and T2-weighted images and fail to enhance with contrast material administration (13). Autosplenectomy appears as a small calcified low-signal-intensity spleen that

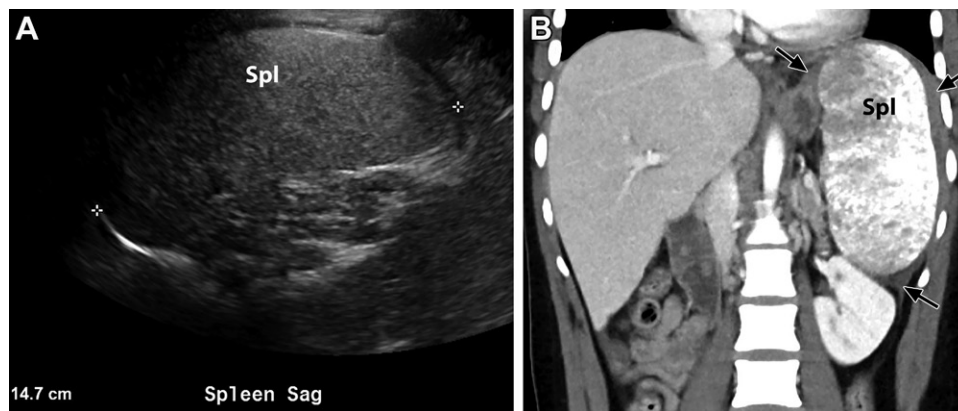


Figure 8. Splenic sequestration and rupture in a 22-year-old woman with SCD who presented with pain and a drop in Hb and hematocrit levels. *Spl* = spleen. (A) Sagittal gray-scale US image shows an enlarged and heterogeneously echogenic spleen measuring 14.5 cm in length. The spleen measured 9.5 cm on US images obtained 2 months earlier (not shown). (B) Correlative coronal contrast-enhanced CT image shows a massively enlarged calcified spleen with hyperattenuating perisplenic free fluid (arrows), consistent with splenic sequestration complicated by rupture. The patient underwent splenectomy to control bleeding.

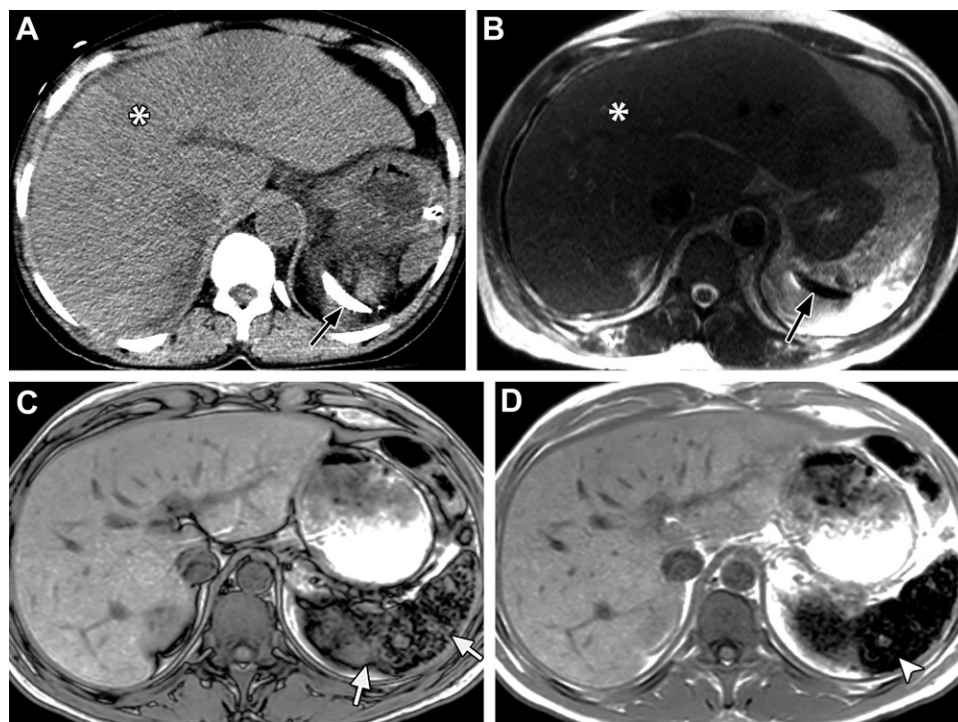


Figure 9. Splenic infarction, autosplenectomy, and fibrosis in two patients. (A, B) Axial noncontrast CT (A) and T2-weighted half-Fourier acquisition single-shot turbo spin-echo (HASTE) MR (B) images of the abdomen in a 47-year-old woman with SCD show a small shrunken spleen that is entirely calcified (arrow in A). Low signal intensity of the spleen is noted (arrow in B). Note the enlarged heterogeneous liver (*). (C, D) In a 23-year-old woman with SCD who presented with abdominal pain, axial out-of-phase T1-weighted MR image (C) shows a small fibrotic spleen with low signal intensity (arrows in C). Axial in-phase T1-weighted MR image (D) shows signal dropout in the spleen (arrowhead in D) consistent with ferrocacinosis and fibrosis.

may contain areas of preserved functional splenic tissue (relatively hyperechoic on US images), fibrosis, or EMH (7,11).

Splenic Abscess

Infarction and superinfection can lead to splenic abscess formation, and patients present with fever

and left upper quadrant pain (7). US images demonstrate a complex hypoechoic unilocular or multiseptated lobulated mass. Depending on the causative organism, air may be seen within the abscess, appearing as multiple echogenic foci with dirty posterior shadowing (7). Peripheral hyperemia may be present.

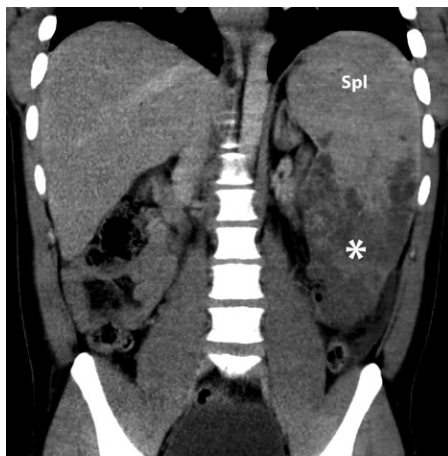


Figure 10. Splenic infarct in a 16-year-old adolescent boy with SCD. Coronal contrast-enhanced CT image of the abdomen and pelvis shows a markedly enlarged spleen (*Spl*) with a large wedge-shaped area of hypoenhancement (*), compatible with an infarct.

Contrast-enhanced CT and MR images show a low-attenuation and T2 hyperintense, respectively, multiseptated collection with irregular lobulated borders and associated edema (7). The enhancement pattern is typically peripheral (whereas active EMH tends to demonstrate minimal-to-mild heterogeneous enhancement) (13,20). Abscess usually shows restriction on diffusion-weighted images (7). Treatment includes antibiotic therapy, percutaneous drainage, or splenectomy (7).

Renal Manifestations

Sickle Cell Nephropathy and End-Stage Renal Disease

Generally, patients with SCD have increased renal plasma flow and glomerular filtration rate from prostaglandin hypersecretion causing vasodilation (19). However, despite increased renal perfusion, medullary perfusion is substantially decreased, leading to VOC in the vasa recta (21). The kidneys are particularly sensitive to microvascular obstruction and ischemia owing to their heavy vascularity and high rate of oxygen consumption (11,22). While the renal cortex is relatively spared from iron deposition in the absence of transfusion therapy, repeat episodes of intravascular hemolysis result in hemosiderin and ferritin accumulation mainly in the proximal convoluted tubules (23,24). In SCD, increased polymerization of HbS in the renal medulla leads to VOC of the vasa recta and subsequent ischemia, infarction, papillary necrosis, and scarring, any of which can progress to acute failure (2,5,11). Not surprisingly, approximately 60% of patients with HbSS have renal involvement (22). Renal disease

Table 2: Characteristics of Sickle Cell Nephropathy

Parameter	Characteristics
Risk factors	Recurrent VOCs of the medulla Sepsis Dehydration Nephrotoxic drugs Multiorgan dysfunction
Presentation	Hyposthenuria Polyuria Enuresis Albuminuria (preceding proteinuria)
Complications	Repeated ischemic insults to renal medulla lead to progressive renal impairment and then end-stage renal disease (ESRD)
Treatment	Avoidance of nephrotoxic medications Angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blocker (ARB) therapy ESRD Recombinant erythropoietin Dialysis Transplantation

Source.—Reference 2.

resulting from SCD is termed *sickle cell nephropathy* (Table 2) (2,11,22,25). Renal vein thrombosis has also been reported, and end-stage renal disease (ESRD) is seen in 10%–15% of patients with HbSS (2,5,11,22).

At US, ESRD may manifest with echogenic cortical parenchyma and loss of corticomedullary differentiation (25,26). At spectral Doppler US, resistive indices are greater than 0.75 (26). The kidneys are atrophic, measuring less than 8 cm in length (27). Delayed nephrogram can be visualized at contrast-enhanced CT. However, intravenous contrast material should be avoided in patients with elevated creatinine levels and low glomerular filtration rate (28). Chronic renal disease is associated with development of small mostly simple renal cortical cysts (22). Loss of corticomedullary differentiation is seen at T1-weighted imaging, and T2-weighted imaging demonstrates low signal intensity of the renal cortex (23,29).

Urinary Tract Infection

Damage to the renal medulla from microvascular obstruction affects the ability of the nephrons to concentrate and acidify urine, resulting in alkaline urine that favors bacterial proliferation and predisposes to urinary tract infection (UTI), including pyelonephritis, cystitis, or renal abscess

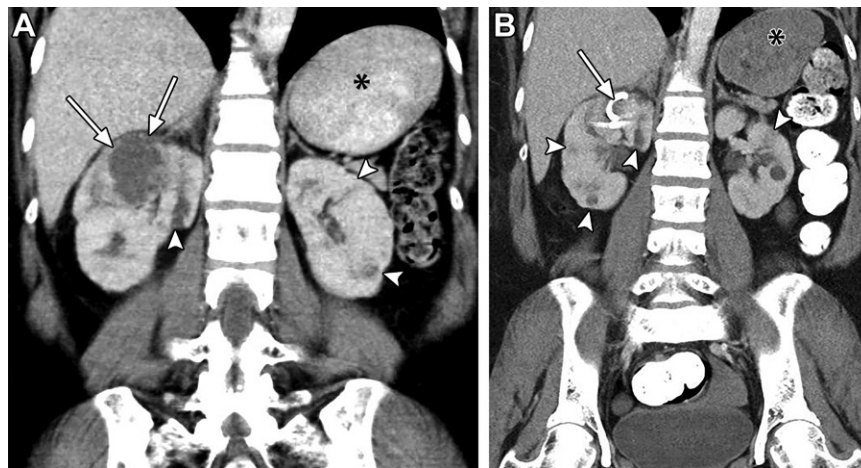


Figure 11. Renal abscess in a 38-year-old woman with SCD who presented with right flank pain and an increased white blood cell count. (A) Coronal contrast-enhanced CT image of the abdomen shows a complex cystic structure in the right upper renal pole (arrows) with mild surrounding inflammation. This finding was not present at a recent prior abdominal CT examination (not shown). The findings are indicative of renal abscess in the setting of pyelonephritis. Note the splenomegaly (*) in A and B). (B) Coronal contrast-enhanced CT image of the abdomen obtained after percutaneous drainage catheter placement (arrow) shows an interval decrease in the size of the upper pole abscess. Note the striated appearance of the renal cortex with areas of hypoattenuation extending through the renal cortex (arrowheads in A and B).



Figure 12. Pyelonephritis in a 34-year-old woman with SCD, urinary tract infection, and costovertebral angle tenderness. Coronal contrast-enhanced CT image shows multiple wedge-shaped areas of low attenuation (arrows) extending through the cortices, giving them a characteristic striated appearance, and mild perinephric stranding. Note the splenomegaly (*).

(Figs 11, 12) (5,30). At CT, pyelonephritis may manifest with a striated nephrogram, but this finding is nonspecific and should be correlated with the clinical history and laboratory findings (Fig 12) (11). Pyelonephritis can become complicated by the development of a renal abscess, which at imaging should be suspected when a new complex cystic area develops in the renal cortex. Hyperemia usually is seen at the periphery of the collection (Fig 11).

Infection can extend to the adjacent soft tissues, resulting in perirenal stranding and/or additional abscesses in the retroperitoneum. Although it can

manifest without imaging findings, cystitis should be suspected if bladder wall thickening and perivesicular inflammatory changes are seen (11).

Renal Cortical Infarcts

In SCD, renal cortical infarcts are characterized by a multifocal bilateral distribution, appearing on contrast-enhanced CT images as wedge-shaped or geographic regions of hypoenhancement with sharp margins (11). Similar to pyelonephritis, multifocal infarction may appear with a striated nephrogram at CT and MRI. Unlike pyelonephritis, cortical rim enhancement is typically preserved owing to a separate arterial supply. This, along with clinical symptoms and laboratory findings, helps to differentiate these two entities (Fig 13) (11). If infarcts are complicated by hemorrhage, perirenal hematoma may be seen (11). Renal cortical necrosis may coexist with renal infarcts and appear as a reversal of the usual corticomedullary pattern, with the cortex being hypointense at MRI relative to the medulla, mimicking hemosiderin deposition (Figs 5, 14).

Renal Papillary Necrosis

In approximately 2.3% of patients with SCD, recurrent VOC leads to ischemia and necrosis of the medulla and renal papillae (2,7). The manifestation of renal papillary necrosis ranges from painless microscopic hematuria to frank anemia, hydronephrosis, and oliguria or anuria if sloughing of the papilla, hemorrhage, and outflow obstruction occur (2,7,11). Early ischemic changes appear

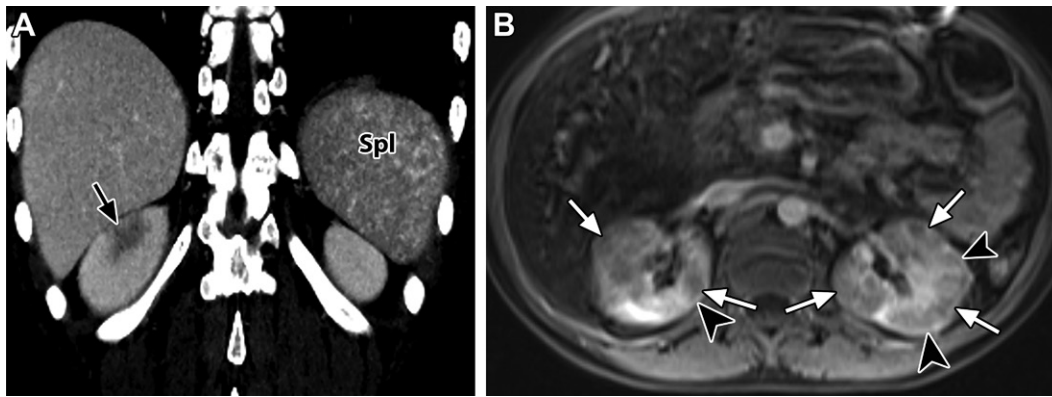


Figure 13. Renal infarct. (A) Coronal contrast-enhanced CT image in a 21-year-old man with SCD shows a wedge-shaped hypoattenuating area in the upper pole of the right kidney (arrow) compatible with an infarct. Note the partially calcified spleen (Spl). (B) Multiple renal infarcts in a 22-year-old man with SCD. Axial gadolinium-enhanced T1-weighted MR image shows multiple nonenhancing areas in the renal cortex (arrows) compatible with renal infarcts, with a characteristic rim of cortical enhancement at the level of the renal infarcts (arrowheads).

as poorly margined hypoenhancing regions at the tips of the medullary pyramids, best appreciated in the nephrographic phase of multiphase CT (31). As ischemia evolves to necrosis, contrast-enhanced CT and MR images demonstrate loss of the normal renal pyramids with distortion of the medullary-calyceal junction and calyceal blunting or clubbing. Small irregular foci of contrast material pooling in necrotic spaces from sloughed papilla are referred to as the “ball-on-tee” and/or “lobster claw” signs (Fig 15) (7,11). Sloughed necrotic papilla can calcify with time. On US images, necrotic renal papillae appear as echogenic foci. Unlike other causes, papillary necrosis in SCD is a bilateral process. Treatments include antibiotic therapy, hydration, and urinary alkalization (7).

Renal Medullary Carcinoma

Renal medullary carcinoma (RMC) is a rare aggressive cancer occurring almost exclusively in patients with sickle cell trait, usually adolescent boys (Fig 16) (7,11). Symptoms at presentation include hematuria, flank pain, a palpable mass, and weight loss (7). Imaging examinations show an infiltrating mass originating from the renal medulla and invading the renal sinus and cortex, often associated with dilatation of the collecting system, perinephric extension, and adenopathy (7,32–34). RMC appears with mixed echogenicity on US images, with loss of renal corticomedullary differentiation and minimal to no flow on color Doppler US images (32,33). On contrast-enhanced CT and MR images, RMC appears as an infiltrating, ill-defined, hypovascular, heterogeneously enhancing central mass that may be accompanied by intraparenchymal satellite nodules (32,35,36). Enhancement tends to be lower than that of the adjacent cortex and medulla, and areas of hemorrhage (high signal intensity on T1-weighted images) and

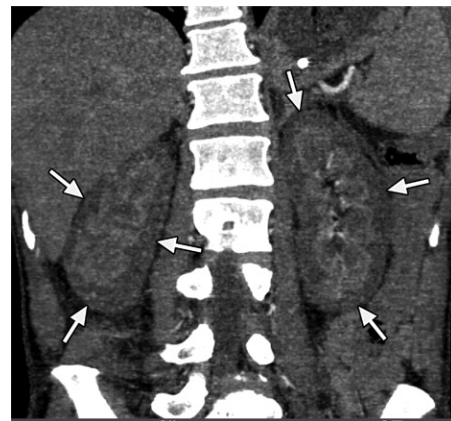


Figure 14. Renal cortical necrosis in a 39-year-old man with renal failure. Coronal CT angiogram of the abdomen shows nonenhancing renal cortices (arrows) compatible with cortical necrosis.

necrosis (high signal intensity on T2-weighted images) are common (33,37,38). RMC restricts diffusion in regions of viable tumor at diffusion-weighted imaging (36,39). Angiography tends to show hypovascularity (36). Treatments include surgery, radiation therapy, chemotherapy, and immune therapy, but the prognosis is poor (7).

Hepatobiliary Manifestations

Hepatomegaly and Hepatic Sequestration

Almost two-thirds of patients develop hepatomegaly owing to acute or chronic sequestration, iron deposition, infarction, or viral infection (7,11). Hepatomegaly is suggested by craniocaudal liver length greater than 16 cm at the midclavicular line, rounding of the inferior liver border, or extension beyond the lower pole of the right kidney (Fig E1) (7).

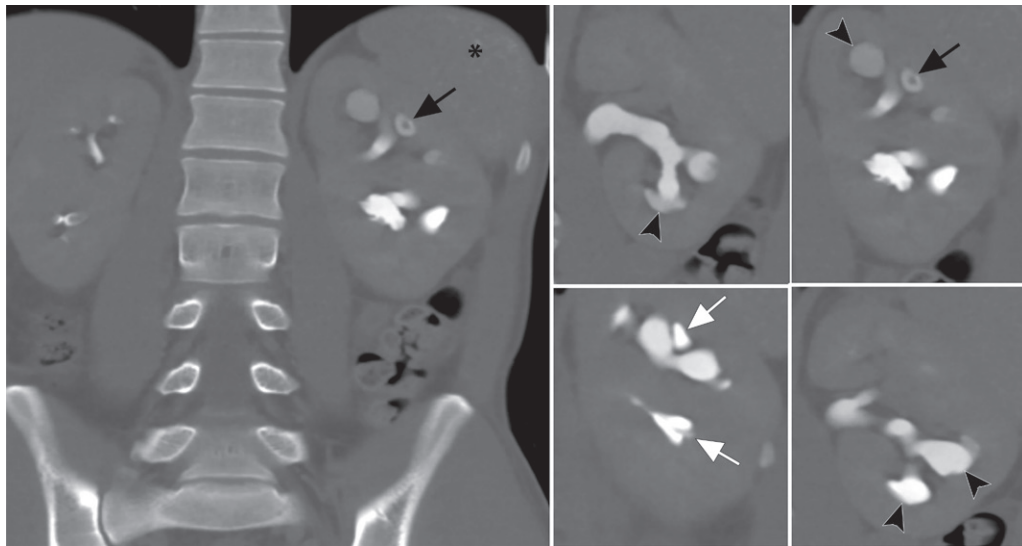


Figure 15. Renal papillary necrosis in a 23-year-old man with SCD who presented with hematuria. At CT urography, the findings are best appreciated in the excretory phase (11,31). Collage of coronal CT urograms with magnifications of the left kidney (right images) obtained in the excretory phase shows abnormal blunted rounded left renal calyces (arrowheads), sloughed papilla (black arrows), and irregular papillae with areas of contrast material pooling in regions of necrosis (white arrows). Note the splenic calcifications (*). The right intrarenal collecting system is intact.

Acute hepatic sequestration is a rare life-threatening complication in which obstruction of the hepatic sinusoids leads to vascular congestion and blood pooling, hypovolemia, and acute anemia (7,11). Patients present with acute right upper quadrant pain and fever (7). Risk factors include acute anemia (including splenic sequestration) and viral and bacterial infections (2). Imaging findings include acute hepatomegaly and differential perfusion (7). Treatment is supportive and can include transfusion (2,7).

Hepatic Infarction and Abscess

Hepatic infarction, occurring in approximately one-third of patients with SCD, manifests with acute right upper quadrant pain and hepatomegaly (11). Over time, fibrosis, capsular retraction, and calcifications may be seen (11). Infarcts appear as rounded or wedge-shaped peripheral areas that are low attenuation on CT images, hypointense on T1-weighted images, and hyperintense on T2-weighted images and demonstrate patchy heterogeneous nonenhancement on CT and MR images (Fig 17) (11). Concern for superinfection and abscess formation should be raised when lobular peripherally enhancing multiseptated collections with adjacent parenchymal edema are detected (7,11).

Cirrhosis and Hepatocellular Carcinoma

Cirrhosis can result from iron deposition from multiple transfusions, hepatitis infection, or repeated infarction (40). Imaging findings include a heterogeneous nodular liver that may be larger than

in those without SCD owing to iron overload (Fig E2) (40). At MRI, associated fibrosis is hypointense at T1-weighted imaging and hyperintense at T2-weighted imaging (40). Decompensated liver cirrhosis can manifest with ascites, varices, splenomegaly, and other signs of portal hypertension (40). At Doppler US, portal hypertension is characterized by sluggish flow with increased respiratory pulsatility (41,42). In advanced disease, dilatation of the portal vein, reversal of flow (hepatofugal flow) in the main vein and its branches, and portal vein thrombosis may occur (41–43).

As in patients without SCD, cirrhosis increases the risk of hepatocellular carcinoma (HCC) in patients with SCD (44). On noncontrast CT images, HCC lesions are typically hypoattenuating and heterogeneous, whereas at contrast-enhanced CT and MRI, HCC demonstrates heterogeneous arterial enhancement with rapid central washout and residual capsular enhancement, although iron overload can confound the appearance of benign and malignant liver lesions (Fig 18) (44). Unlike hypointense regenerating nodules, HCC tends to be hyperintense relative to the liver at T2-weighted imaging (Fig 18) (44). While restricted diffusion is indicative of malignancy, it is a nonspecific imaging feature (45,46).

Imaging features more specific for HCC include intralesional fat (with signal dropout on opposed-phase images compared with on in-phase gradient echo T1-weighted images), corona enhancement (transient enhancement of the venous drainage portion of the peritumoral parenchyma at CT or MRI), nodule-in-nodule architecture

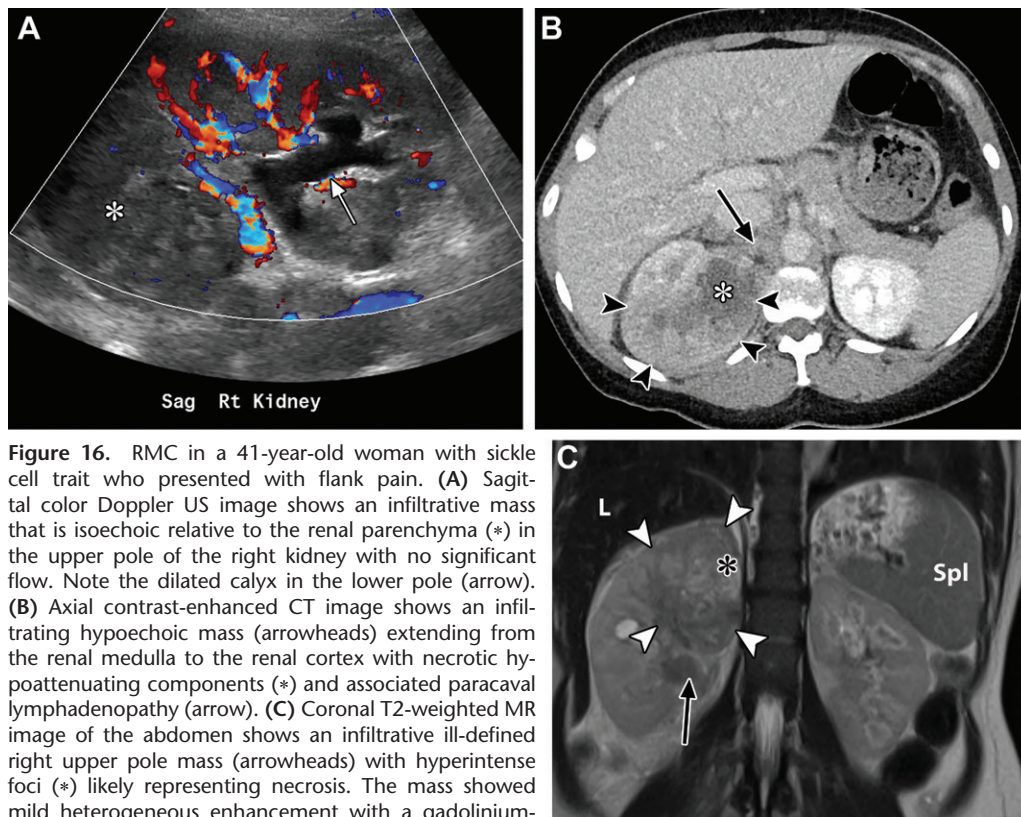


Figure 16. RMC in a 41-year-old woman with sickle cell trait who presented with flank pain. (A) Sagittal color Doppler US image shows an infiltrative mass that is isoechoic relative to the renal parenchyma (*) in the upper pole of the right kidney with no significant flow. Note the dilated calyx in the lower pole (arrow). (B) Axial contrast-enhanced CT image shows an infiltrating hypoechoic mass (arrowheads) extending from the renal medulla to the renal cortex with necrotic hypoattenuating components (*) and associated paracaval lymphadenopathy (arrow). (C) Coronal T2-weighted MR image of the abdomen shows an infiltrative ill-defined right upper pole mass (arrowheads) with hyperintense foci (*) likely representing necrosis. The mass showed mild heterogeneous enhancement with a gadolinium-based contrast material sequence (not shown). There is dilatation of the renal pelvis containing debris and likely blood products (arrow). Note the low signal intensity in the liver (L) indicative of transfusion-related iron overload. The spleen (Spl) is enlarged. Follow-up imaging performed 6 months later (not shown) showed interval growth of the mass with extension into the adjacent pararenal fat and liver, with significant paracaval lymphadenopathy and hepatic and lung metastases.

(nodule within a larger nodule or mass, uncommonly seen at CT or MRI), or mosaic architecture (cluster of nodules and/or compartments with variable shape, size, enhancement, and signal intensity) (45–47).

Viral Hepatitis Infection

There is a higher incidence of hepatitis B and C infection in SCD secondary to multiple blood transfusions, although universal screening of blood products in the United States has led to reduced prevalence of hepatitis C infection in patient cohorts who underwent transfusion after 1992 (48). Patients with viral hepatitis infection present with fever, hepatomegaly, and jaundice (7).

At US, hyperechoic biliary triads on a background of hypoechoic edematous liver parenchyma create a “starry sky” appearance (Fig 19) (7). Periportal edema is depicted as hyperechogenicity or cuffings tracking on both sides of the portal triads, corresponding to a similar pattern of areas of hypoattenuation depicted at CT and hyperintensities at T2-weighted imaging, which helps to differentiate periportal edema from biliary ductal dilatation (49,50).

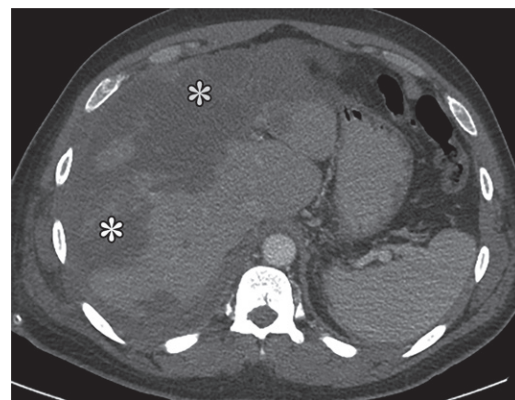


Figure 17. Hepatic infarct in a 40-year-old man with SCD who presented with severe acute abdominal pain. Axial contrast-enhanced CT image shows an extensive area of hypoattenuation (*) in the right hepatic lobe, indicative of infarction.

Biliary Disease

Biliary disease in SCD ranges from benign sludge formation to frank liver failure (51). More than 85% of patients with SCD are diagnosed with black pigment gallstones composed of calcium bilirubinate, formed owing to chronic hemolysis and hemoglobin incorporation into bile (7,11,52,53). Gallstones may be found incidentally or manifest

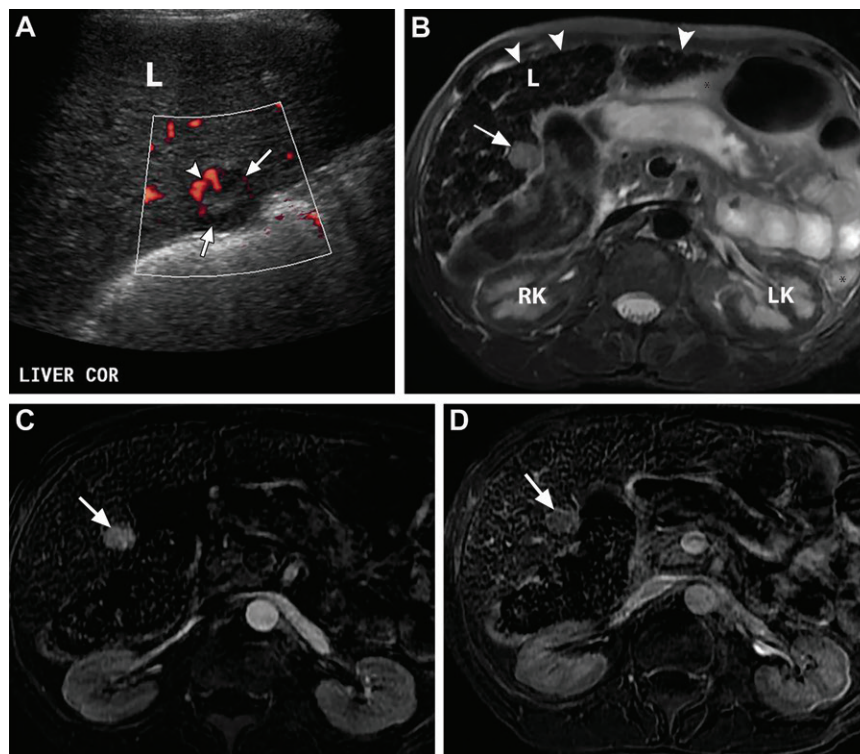


Figure 18. HCC in a 54-year-old man with SCD, hepatitis C virus infection, and cirrhosis. (A) Coronal power Doppler US image shows coarse heterogeneous liver (L) parenchyma. A 2-cm hypoechoic mass (arrows) is seen at the periphery of the right hepatic lobe that demonstrates flow (arrowhead), a finding concerning for HCC. (B) Correlative axial T2-weighted MR image shows a hepatic mass (arrow) at the periphery of the right hepatic lobe. The mass is hyperintense compared to the background liver (L). Note the nodularity of the liver contour (arrowheads), indicative of liver cirrhosis. There is marked hypointensity of the liver and bilateral renal cortices (right kidney [RK], left kidney [LK]) compatible with iron deposition. Note the small amount of ascites (*) likely related to decompensated cirrhosis and portal hypertension. (C, D) Corresponding arterial and portal venous subtraction MR images show arterial enhancement of the hepatic lesion (arrow in C) and portal venous washout (arrow in D), which help confirm the diagnosis of HCC.

with abdominal pain from biliary colic, cholecystitis, choledocholithiasis, cholangitis, or gallstone pancreatitis, with biliary colic occurring in about two-thirds of patients (Figs E4, E5) (7,11). While data on cholecystitis in SCD are limited, one study showed that laparoscopic cholecystectomy helped confirm the diagnosis in 136 of 496 patients with SCD (27.4%) (54). On US images, pigment stones are echogenic mobile structures in the gallbladder with posterior shadowing (Fig E4) (7). At CT, they are radiopaque in 83% of cases and radiolucent in 17% of cases, limiting visualization (7,55). On MR images, pigment stones show increased T1 signal intensity (unlike cholesterol stones), although they are best visualized at T2-weighted imaging and steady-state imaging, demonstrating low T2-weighted signal intensity (7,56). In cholecystitis, gallbladder wall thickening and inflammation may be seen (7).

At technetium 99m (^{99m}Tc) diisopropyl-imino-diacetic acid imaging, prolonged nonvisualization of the gallbladder is suggestive of acute chole-

cystitis, while delayed visualization is suggestive of chronic cholecystitis (51). Treatment often includes elective or emergent cholecystectomy (7).

Hepatopathy and Cholestasis

Hepatopathy refers to hepatic dysfunction often associated with cirrhosis and elevated bilirubin levels not explained by acute symptoms and can coincide with cholestasis or sequestration (7,51).

Cholestasis and cholangiopathy occur when recurrent end-arterial and hepatic sinusoidal obstruction results in ischemic damage to the biliary tree and hepatotoxicity from disruption of biliary flow (7,11,51). While rare, acute presentations—typically including fever, abdominal pain, and increasing bilirubin levels—are clinical emergencies (7). Initially, images demonstrate dilated biliary ducts (Fig E5), with recurrent insult leading to intra- and extrahepatic biliary duct strictures or biloma formation (Fig 20, Movie) (7,11,51,57). Although treatment is mainly supportive, liver failure may occur and require transplantation (7).

Pancreatitis

Limited data on pancreatitis in patients with SCD suggest a prevalence similar to that of the general population (7,58). A few case studies of SCD-associated pancreatitis feature patients of all ages and those with coinheritance of β -thalassemia (59–63). Pancreatitis can occur when VOC results in ischemia or secondary to obstruction by biliary stones (7,60). Clinical presentation includes fever, nausea and/or vomiting, and epigastric pain radiating to the back (7). At contrast-enhanced CT and MRI, uncomplicated pancreatitis appears with mild pancreatic enlargement and edema with or without peripancreatic inflammatory changes, and decreased T1 signal intensity and parenchymal enhancement (Fig 21) (7,11). In necrotizing pancreatitis, nonenhancing pancreatic parenchyma can be seen (11).

Potential complications include peripancreatic walled-off necrosis, pseudocyst formation, pseudoaneurysm, or venous thrombosis (11,62). Treatment depends on the cause but mainly includes fluid resuscitation, early enteral feeding, and pain control. In gallstone pancreatitis, cholecystectomy may be required (7,11).

Gastrointestinal Manifestations: Bowel Ischemia and Sequelae

VOC of the small bowel and small bowel mesentery often develops in SCD and can result in ileus, ischemic enteritis and colitis, bowel infarction, necrosis, and even perforation (7,11). CT images may demonstrate segmental or diffuse bowel wall thickening (7,11). The “target” sign, while a nonspecific feature, may be seen on contrast-enhanced CT images owing to alternating contrast enhancement of hyperemic mucosa, hypoattenuation of edematous submucosa, and enhancing hyperemic serosa (11,64,65). On noncontrast CT images, hypoattenuating submucosal edema versus underlying mucosal hemorrhage may appear with a pseudo-target sign (11). Hypoenhancement of mucosa or air within the bowel wall (pneumatosis intestinalis) or mesenteric/portal venous gas is suggestive of necrosis (Fig 22) (7,11). Superinfection and perforation of ischemic bowel can lead to abscess formation, while chronic ischemia can lead to gastric ulceration and peptic ulcer disease (11). Treatment of ischemic bowel frequently requires surgical resection (7).

Genital Manifestations: Priapism and Atrophic Testes

SCD is the most common cause of priapism, with a 30%–40% lifetime risk in patients with SCD, and is the most common cause of genital pain in children. It is characterized by prolonged (>4 hours) painful penile erection unrelated to

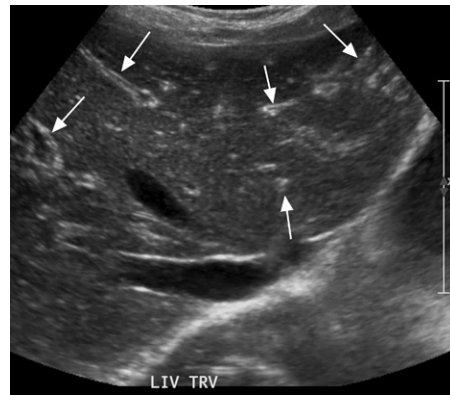


Figure 19. Viral hepatitis infection in an 8-year-old boy with SCD who presented with jaundice and tender hepatomegaly. Gray-scale abdominal US image shows an enlarged hypoechoic liver with a starry-sky appearance owing to edema of the portal triads (arrows).

sexual activity, with ischemia occurring secondary to low-flow VOC (2,7,66). While the corpus spongiosum in patients without SCD is typically not involved and the glans remains soft, priapism in SCD is ischemic in cause and demonstrates tricorporal involvement (66). US images show corpus cavernosum and/or spongiosum thrombosis with decreased to absent flow in the cavernosal and spongiosal arteries at color Doppler US (7). CT images demonstrate an erect penis, and MR images show increased T1 signal intensity in the penile corpora. Ischemic corporal tissue fails to enhance with intravenous contrast material (Fig 23) (7,66). Treatment includes hydration, surgical decompression, sympathomimetic injection, and pain management, with simple transfusion or erythroexchange (blood is removed and replaced) if surgical intervention or penile decompression is delayed (2,7).

Almost a quarter of men with SCD develop atrophic testes and hypogonadism secondary to VOC of the testicular vessels (with or without hydroxyurea) (7). US, CT, and MR images depict small testicular size and fibrotic changes (7). The treatment is testosterone hormone replacement therapy.

Pregnancy Complications

Pregnancy-associated mortality rates in women with SCD range from six to 22 times higher than those in women without SCD (2,8). Patients are at increased risk for hypertension, preeclampsia, miscarriage, maternal death, preterm delivery, fetal intrauterine growth retardation, urinary tract infections, pyelonephritis, thromboembolism, VOCs, ACS, and hemolytic crises (2). Management is multidisciplinary and dependent on close observation throughout pregnancy and the postpartum period (2).

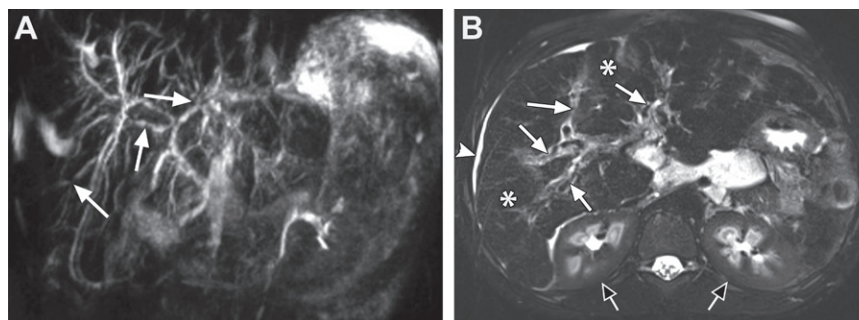


Figure 20. Intrahepatic cholestasis and bile plugs in a 9-year-old boy with SCD and elevated bilirubin and liver enzyme levels. (A) Coronal MR cholangiopancreatographic maximum intensity projection after cholecystectomy shows areas of segmental dilatation and narrowing throughout the intrahepatic biliary tree (white arrows). (B) Axial T2-weighted fat-suppressed MR image of the abdomen and pelvis shows markedly low signal intensity in the liver (*) with associated hepatic contour nodularity indicative of hepatic fibrosis and/or cirrhosis. Extensive periportal edema is evident by the hyperintense signal along both sides of the portal triads (white arrows) consistent with hepatopathy. There is irregularity of the intrahepatic biliary ducts with focal areas of stricture. Iron overload is also reflected in the low signal intensity of the renal cortex (black arrows). Trace ascites (white arrowhead) are also depicted.

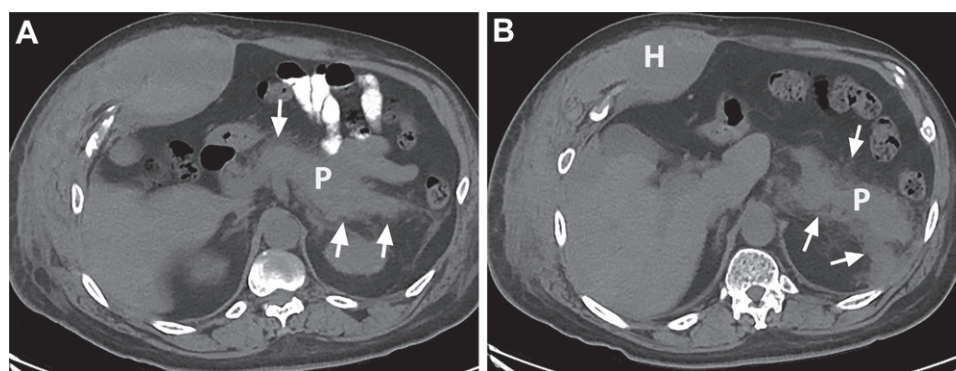


Figure 21. Pancreatitis in a 62-year-old man with SCD who presented with back pain. Axial noncontrast CT images of the abdomen show an edematous pancreas (P) with marked peripancreatic inflammatory changes along the body (arrows in A) and tail (arrows in B) of the pancreas and a peripancreatic fluid collection (not shown) at the head-neck junction, indicative of pancreatitis. Note the large right rectus abdominis muscle hematoma (H) owing to prior anticoagulation therapy.

Other Manifestations of SCD at Abdominal and Pelvic Imaging

Although musculoskeletal and pulmonary systems are only partially included in abdominal and pelvic imaging, knowledge of the manifestations of SCD in these systems is essential to avoid missed diagnoses. In this section, we provide a brief review of the most commonly seen musculoskeletal and pulmonary complications of SCD that can be encountered at abdominal and pelvic cross-sectional imaging.

Musculoskeletal Manifestations

Intramedullary Hematopoiesis.—During normal development, red marrow in the appendicular skeleton converts to yellow marrow over time, but in SCD, reversion may occur (Fig E6) (7). At imaging, diffuse trabecular coarsening can be seen with widening of medullary spaces

and thinning of cortical bone (17). In the spine, bony changes result in biconcave deformity of vertebral bodies from compression by intervertebral discs, with vertebral collapse potentially leading to kyphosis (17). At MRI, intramedullary hematopoiesis is iso- to hypointense to the disk at T1-weighted imaging; has intermediate to low signal intensity at T2-weighted imaging; has similar signal intensity (but is not hyperintense) to relative muscle on the short τ inversion-recovery sequence; and is mildly enhancing with gadolinium-based contrast material (67).

Bone Infarction and Osteonecrosis.—Bone infarction results from sickling of RBCs in the bone marrow, with resultant ischemia and tissue hypoxia further exacerbating sickling (2,7,17). Infarction most commonly occurs in medullary cavities and epiphyses of the bones, often demonstrates serpiginous regions of sclerosis, and

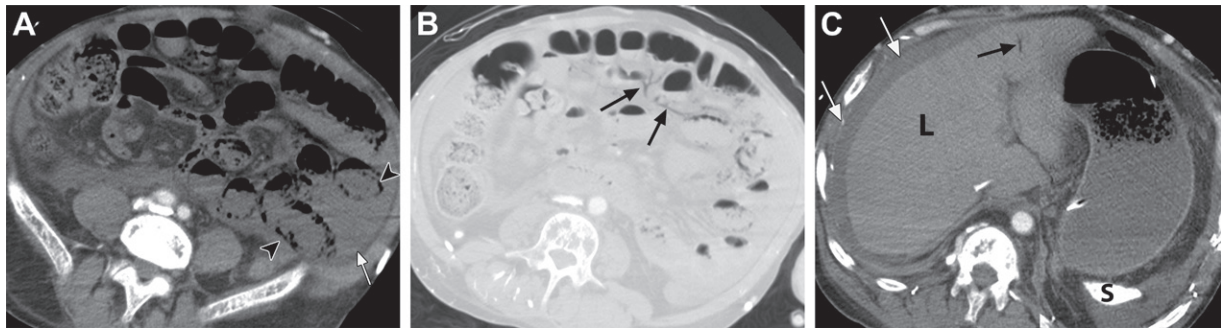


Figure 22. Diffuse bowel infarction in a 35-year-old patient with SCD who presented with a pain crisis. Axial contrast-enhanced CT images of the abdomen show loss of bowel wall enhancement with extensive pneumatosis intestinalis (arrowheads in A) throughout the jejunum and proximal ileum. There is associated mesenteric gas (arrows in B) and a small amount of portal venous gas (black arrow in C). A small-to-moderate amount of free fluid is present (white arrows in A and C). Note the diffuse sclerosis of the pelvic bones. Note the liver (L) enlargement and calcified spleen (S) after autosplenectomy. A partially visualized feeding tube is seen in the stomach.

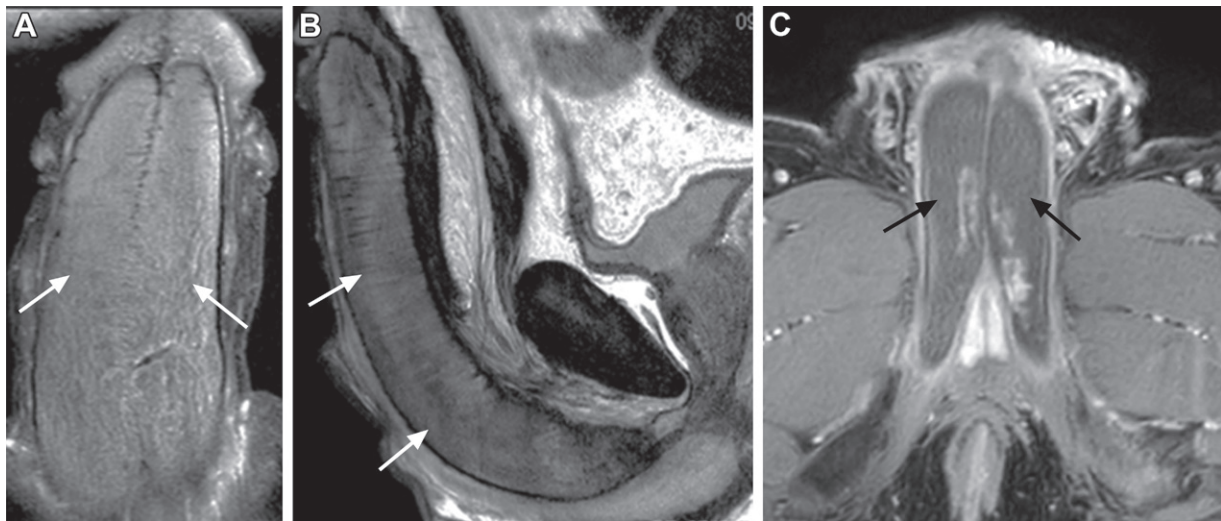


Figure 23. Priapism in a 36-year-old man with SCD. (A, B) Coronal T1-weighted (A) and sagittal T2-weighted (B) MR images of the penis show bilaterally enlarged cavernosa that are hyperintense on the T1-weighted image (arrows in A) and hypointense on the T2-weighted image (arrows in B). The penis is in the erectile position. (C) Coronal gadolinium-enhanced T1-weighted MR image shows no enhancement of both corpora cavernosa (arrows), helping to confirm the ischemic nature of priapism. (Case courtesy of Christine Cooky Menias, MD, Mayo Clinic Arizona, Phoenix, Az.)

can be associated with subperiosteal hemorrhage (Fig E7) (17). Central growth plate infarction in the spine can result in endplate depressions of the vertebral bodies, resulting in H-shaped vertebrae and adjacent “tower vertebrae” with visible growth deformities (Fig 24) (17). Radiographs and CT images show bony sclerosis owing to medullary infarction and resultant dystrophic calcification (7,17). MRI is performed to help verify the presence of and monitor bony infarction (2,17).

Osteonecrosis commonly occurs in patients with SCD and often causes collapse of the articular surfaces of the bones (most commonly the femoral head or shoulder) (2,17). Osteonecrosis manifests with joint pain and limited movement and tends to be bilateral in SCD (Fig E8) (17). Approximately 50% of patients develop osteonecrosis by 35 years of age, although it is less com-

mon in children, likely owing to the nutritional role of synovial fluid for the epiphysis (17). While plain radiographs may initially be normal, over time the epiphysis becomes more heterogeneous, developing crescent-shaped subchondral lucency and eventually depression of the articular surface with collapse and fragmentation (17). These changes facilitate further degeneration (including osteophyte formation) especially at weight-bearing joints (17). CT similarly demonstrates signs of osteopenia and progressive microfractures eventually leading to sclerosis, cortical collapse, and fragmentation (7).

On T1-weighted images, osteonecrosis appears with linear low signal intensity (7). T2-weighted images may show a “double-line sign” from low peripheral signal intensity associated with reactive bone interface adjacent to the inner high-T2-signal-intensity lines associated

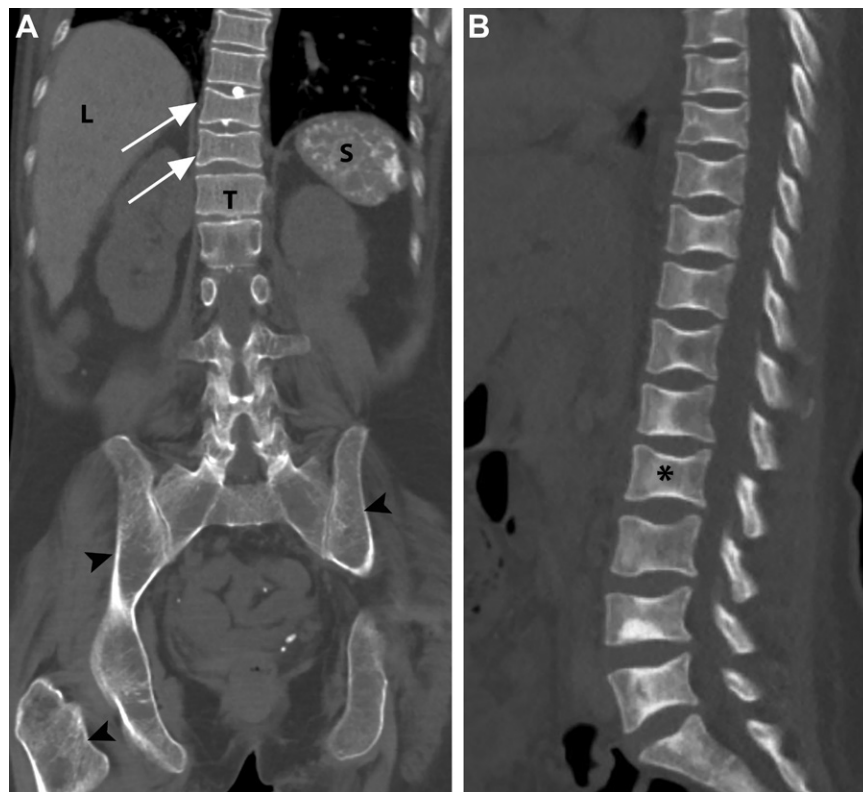


Figure 24. H-shaped vertebrae in a 26-year-old woman with HbS/ β^0 -thalassemia. Coronal (A) and sagittal (B) CT images of the thoracolumbar spine and pelvis show multilevel H-shaped vertebrae, characterized by vertebral endplate depressions (arrows in A, * in B) and diffuse osteopenia (arrowheads in A). The H-shaped vertebrae may be accompanied by tower vertebrae (T), defined as compensatory lengthening of vertebrae adjacent to H-shaped vertebrae (17). Note diffuse osteosclerosis. Characteristic spleen calcifications are depicted (S), and the liver (L) is enlarged and has a cirrhotic appearance.

with the inflammatory response from granulation tissue formation (Figs E7, E9) (7,17). On inversion-recovery T2-weighted images, bone marrow edema appears as regions of high signal intensity (17). Treatment is conservative, although core decompression or joint replacement may be required (2,7).

Osteomyelitis, Osteopenia, and Osteoporosis.—

Tissue and bone infarction, splenic dysfunction, and iron deposition in patients with SCD increase the risk of osteomyelitis, which affects 18% of patients (5,7,17). While long bone diaphyseal infections are most frequent, any bone may be affected and hematogenous spread may lead to multifocal infection (2,5,17). The patient presentation includes pain, fever, and tumescence (17). Imaging findings of early osteomyelitis at CT and US are often nonspecific, and MRI is the modality of choice.

Bone marrow changes are seen within 3–5 days as hyperintensity on fat-saturated T2-weighted images owing to edema. Soft-tissue infection extension, joint effusion, sinus tract, and abscess formation can also be seen (2,68). The demonstration of hyperintense intramedullary fat globules at T1-weighted imaging is pathognomonic for acute osteomyelitis (68). T1-weighted images may also demonstrate hypointense fluid, although the presence of hematopoietic marrow can be confounding (7,17).

Gadolinium-enhanced T1-weighted images may demonstrate irregular peripheral bone marrow enhancement surrounding a region of nonenhancement (17). While bone marrow edema; adjacent soft-tissue fluid collections; and abnormal periosteal, muscle, fascia, or fat enhancement with gadolinium are nonspecific findings common to both osteomyelitis and infarction, a combination of fat-saturated T2-weighted imaging and intravenous gadolinium-enhanced T1-weighted imaging, as well as imaging features including the location of the abnormality or the presence of serpiginous borders or a double-line sign, can help differentiate infarction from osteomyelitis (17). Nuclear imaging examinations are helpful in differentiation of osteomyelitis from other pathologic conditions.

Approximately 7% of patients develop septic arthritis, typically co-occurring with VOC and bone infarction (17). If acute, large joint effusion may be depicted at radiography, CT, and US, with MR images showing perisynovial edema and synovial enhancement after intravenous gadolinium administration (Figs E10, E11) (17). Antibiotic therapy may be combined with surgical débridement/drainage for osteomyelitis and fluid aspiration for septic arthritis (7).

Osteopenia and osteoporosis occur owing to the high prevalence of vitamin D deficiency in SCD and contribute to vertebral fragility and development of compression fractures (2,17).

Table 3: Pulmonary Manifestations of SCD

Manifestation	Pathophysiology	Imaging Findings	Management
Pneumonia	Chronic vaso-occlusion and repeat infarction lead to splenic dysfunction with disruption of opsonization and the complement cascade, ultimately resulting in increased susceptibility to infection with encapsulated organisms	Radiography: consolidation CT: consolidation, ground-glass opacification, and/or tree-in-bud nodularity	Prophylaxis: immunizations* Treatment: antibiotic therapy, with <i>Streptococcus pneumoniae</i> and gram-negative enteric organism coverage in children
ACS [†]	Vaso-occlusion of pulmonary microvasculature Triggers and/or risk factors: infection, infarction, hypoventilation and/or asthma, prior ACS event, abdominal surgery, VOC-induced mesenteric ischemia	Radiography: lower lobe–predominant opacities CT and MRI: lower lobe–predominant opacities in mosaic pattern	Antibiotic therapy, oxygen, hydration, anticoagulation therapy, bronchodilators, blood transfusion, erythroexchange, pain management
Pulmonary embolism, infarction	Hypercoagulability, secondary to dysfunctional fibrinolysis; intravascular hemolysis leads to release of procoagulant factors, which increase platelet function and RBC sickling, leading to disruption of blood flow; chronic catheterization Bone infarction, which can result in fat embolism	Embolism at CT and MRI: pulmonary arterial filling defects Infarction at radiography, CT, and MRI: peripheral wedge-shaped consolidation	Anticoagulation therapy
Chronic fibrosis, interstitial lung disease	Repeat VOCs lead to chronic fibrosis	Radiography, CT, and MRI: scarring, fibrosis, and volume loss	Supportive
Pulmonary hypertension	Chronic vascular congestion with progressive reduction in pulmonary vascular flow results in recurrent VOCs Chronic hemolysis results in the release of free hemoglobin, which neutralizes nitric oxide and results in catalysis of free radicals	CT and MRI: main pulmonary artery diameter greater than 29 mm, right heart enlargement	Prevention: medical SCD therapy, management of triggers, risk factor modification Treatment: prostacyclin pathway agonists, endothelin receptor antagonists, nitric oxide enhancers

Sources.—References 2,5–8,69,70.

*Immunizations with pneumococcal and *Haemophilus influenzae* vaccines are vital for patients with autosplenectomy owing to their predisposition to infection with encapsulated bacteria (6). Infections are treated with antibiotic therapy, with *S. pneumoniae* and gram-negative enteric organism coverage in children (2).

[†]In severe cases, ACS can progress to acute respiratory distress syndrome (69).

Pulmonary Manifestations

Pulmonary manifestations can be detected at abdominal CT owing to inclusion of the lower portion of the chest (Table 3) (2,3,5–8,69,70). Infection is one of the most common triggers of ACS, a leading cause of hospitalization and death in patients of all ages (Figs 25, E12) (2,3,5,7,69,71). ACS can also be triggered by pulmonary embolism and infarction, with a hypercoagulable state increasing the risk of venous thromboembolism by 3.5-fold (7,70). In severe cases, ACS can progress to acute respiratory distress syndrome and chronic lung disease (Fig E13) (7,69).

Mainstays of Management and Treatment of SCD

In patients with SCD, blood transfusions are the mainstay of treatment of acute anemia by both increasing the oxygen-carrying capacity of blood and reducing complications of VOC by decreasing the amount of circulating HbSS (72). Newborn screening, immunizations, improved detection and treatment of infection with prophylactic and therapeutic antibiotic therapy, and disease-modifying medications have drastically improved patient survival (6). In the United States, mortality rates in children younger than 3 years have declined by 68%

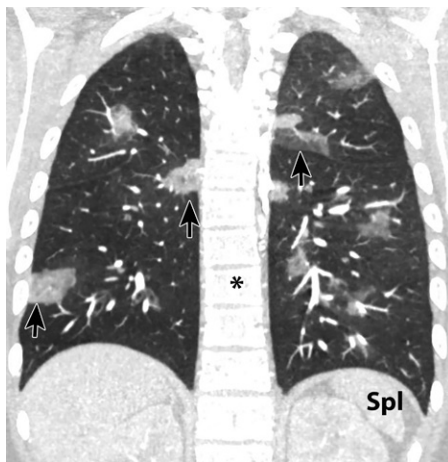


Figure 25. COVID-19 pneumonia in a 16-year-old girl who presented with shortness of breath and was diagnosed with ACS. Coronal CT angiogram of the chest shows patchy bilateral peripheral ground-glass opacities (arrows) indicative of typical COVID-19 pneumonia. Note small spleen (Spl) owing to autosplenectomy and mildly H-shaped vertebrae (*).

over the past 2 decades, with 94% surviving to adulthood (6). Environmental and socioeconomic factors are influential: in regions without comprehensive care, death in early childhood remains common and treatment is expensive and therefore may be inaccessible (2,3,6). Even with the best medical care, life expectancy remains approximately 30 years less than in nonaffected patients (3,6).

Conclusion

Abdominal and pelvic manifestations of SCD are primarily due to VOC, hemolysis, or infection. Cross-sectional imaging with US, CT, and MRI plays an essential role in the identification of complications of SCD. The affected organs are variably involved. Prompt diagnosis and management depend on recognition of early complications. Advancements in technology and new treatment development will continue to improve understanding and management of this complex disease.

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